COOX

Data Analytics Project

Non-Serviceable Bookings Analysis

& Geospatial Blocking Strategy

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1. Executive Summary

This project analyzes COOX's historical booking data to identify geographical patterns of non-serviceable locations i.e. areas where delivery services cannot be fulfilled, resulting in refunds. The primary objective was to detect high-frequency clusters of these failed bookings and generate an actionable list of pincodes to block, thereby reducing revenue leakage and improving overall customer experience.

Key outcomes of the analysis:

- Applied DBSCAN geospatial clustering with a precision of 2-km radius.
- Identified 53 distinct problem clusters predominantly located on the outskirts of major metropolitan areas.
- Deliberately filtered out 300+ isolated booking failures as statistical noise, thus protecting legitimate serviceable areas.
- Delivered a non-duplicated, prioritised pincode blocklist ready for immediate operational deployment.

2. Problem Statement & Objectives

Overview

At COOX, bookings are frequently received from locations outside the current serviceable areas , typically city outskirts. Because these bookings cannot be fulfilled, they must be cancelled and refunded. This recurring pattern leads to compounding operational and business issues:

- Revenue leakage and refund processing overhead.
- Wasted operational time and logistics resources.
- Poor customer experience for users who complete payment but receive a cancellation.

Project Objectives

The core objectives of this project, derived directly from the provided problem statement, were:

- Analyze Distribution: Identify city-wise trends and detect frequently occurring non-serviceable areas.
- Geospatial Mapping: Use latitude/longitude data to create interactive visualisations (heatmaps and cluster maps).
- Cluster Identification: Apply machine learning clustering to detect high-density booking zones and tag them as 'Outskirts Hotspots.'
- Pincode Extraction: Extract the corresponding 6-digit Pincodes for all identified clusters via reverse geocoding.
- Blocking Recommendation: Provide a finalized, data-backed list of Pincodes that should be blocked at checkout.

3. Methodology

The project was executed through a structured, end-to-end data pipeline using Python within a Jupyter Notebook environment.

Step 1 - Data Loading and Validation

The analysis began by loading the dataset of 557 refunded bookings. Each record contained a Booking ID, Area Name, City, State, and geographic coordinates (Latitude/Longitude).

- Verified that all records carried a 'refunded' status to ensure data integrity.
- Identified and removed a minor subset of records (1.97%) that were missing latitude/longitude values.

Step 2 - Exploratory Data Analysis (EDA)

Before applying machine learning, the data was visualised to understand baseline distribution patterns across cities, states, and address types. This revealed:

- City & State Distribution: The highest volume of non-serviceable bookings originated from Maharashtra, Karnataka, and Telangana specifically Pune, Bengaluru, and Hyderabad.
- Address Type Breakdown: Alongside Home addresses, Farmhouse was a heavily recurring category for cancelled orders, pointing to a distinct demand segment in rural adjacent areas.

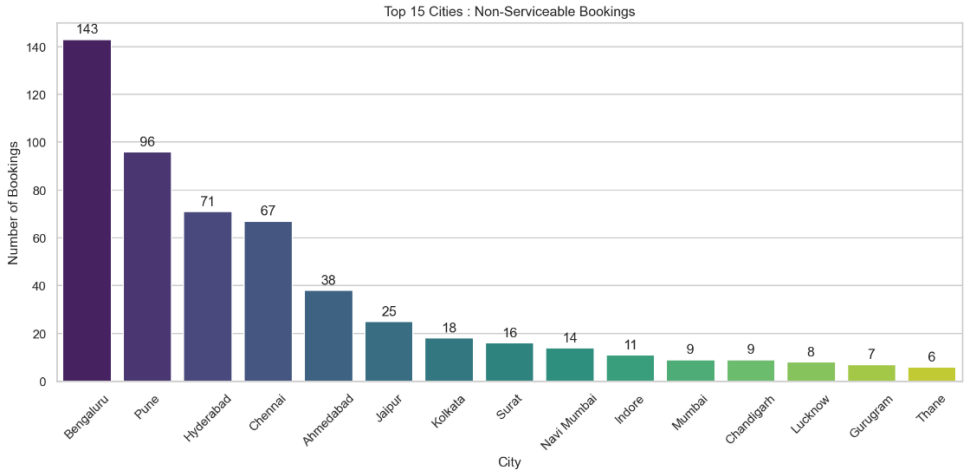

Figure 1: City-wise distribution of non-serviceable bookings

Step 3 - Geospatial Heatmap Creation

Using the Folium library, an interactive heatmap was generated to display all valid booking coordinates across India. This visualisation provided an immediate macro level view of where non-serviceable bookings were concentrated , forming the basis for targeted clustering in the next step.

Step 4 - DBSCAN Clustering (Machine Learning)

To identify systematic 'Hotspots' rather than loose scatter points, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm was applied. DBSCAN was selected over K-Means because it can discover clusters of arbitrary shapes and automatically isolates outliers as noise which is a critical requirement for geographic data.

Key Algorithm Parameters

Parameter	Value	Rationale
eps (Radius)	**2 km**	Matches typical Indian urban pincode footprint (~15-20 sq km). Prevents incorrectly merging entire cities.
min_samples	**3**	Requires at least 3 failed bookings in proximity before flagging a systematic cluster.
Result	**53 Clusters**	300+ isolated noise points excluded, protecting legitimate serviceable areas from accidental blocking.

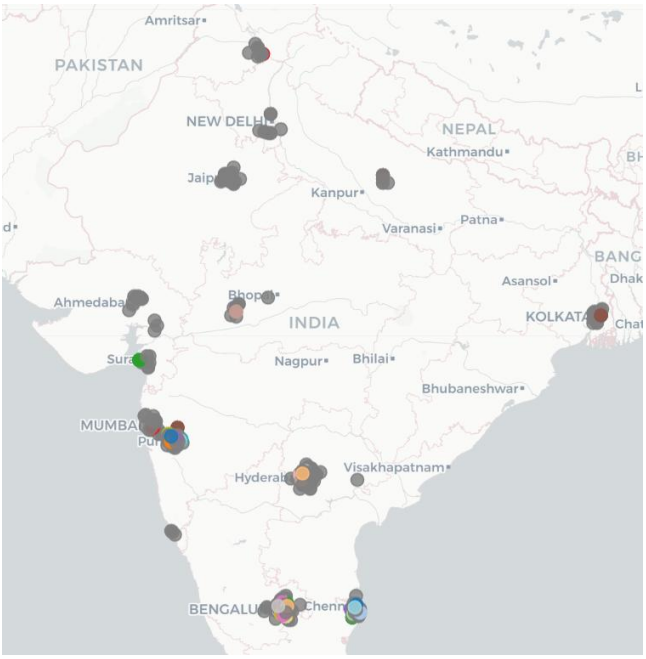

Figure 2: DBSCAN cluster map showing 53 identified hotspots across India

Step 5 - Cluster Boundary Generation

For each of the 53 identified clusters, geometric boundaries were generated to precisely define the service gap zones. Using the Shapely library, Convex Hull polygons were drawn around the outermost coordinates of each cluster , effectively creating “No-Go Zones” on the geospatial map for operational reference.

Step 6 - Pincode Extraction via Reverse Geocoding

To translate coordinate clusters into operationally actionable data for the engineering team, reverse geocoding was performed in three sub-steps:

- The exact centroid (center point) of each cluster was calculated.
- Centroids were passed through the Nominatim API (via Geopy) to extract real-world 6-digit Pincodes, City names, and States.
- A rate-limiting delay (time.sleep) was used to respect API request limits during bulk extraction.

Step 7 -Blocklist Deduplication and Prioritisation

Because multiple 2 km hotspots can occasionally overlap within a single large postal boundary, the extracted data was grouped and deduplicated by Pincode. The final list was sorted in descending order by the total number of failed bookings, ensuring the most problematic areas are addressed first.

4. Key Findings & Insights

Finding 1 The Peri-Urban Scenario

The majority of non-serviceable bookings are highly concentrated on the immediate outskirts of major metropolitan areas particularly the Pune-Mumbai corridor, Bengaluru, and Chennai. The clusters indicate that significant latent demand exists just a few kilometres beyond the current operational delivery limits, representing both a risk (current refunds) and an opportunity (potential expansion zones).

Finding 2 The Farmhouse Trend

Farmhouses represent a significant and distinct portion of non-serviceable bookings. Being leisure properties located in rural-adjacent areas, they generate high demand but sit entirely outside traditional hyper-local delivery zones. This constitutes a recurring, predictable segment of unmet demand that warrants a dedicated business strategy.

Finding 3 High-Precision Blocking is Required

By filtering out noise points, the data clearly shows that broad regional blocking is both unnecessary and harmful to revenue. Systematic non-serviceability occurs in tight, specific 2 km pockets , not across entire cities or districts. This makes pincode-level blocking the only appropriate and responsible intervention.

Finding 4 Top Problem Pincodes by Booking Failures

The non duplicated blocklist highlights the five most severe areas driving systematic refunds:

Rank	Pincode	Location	State	Cluster Count
1	654321	Pimpri-Chinchwad	Maharashtra	Highest
2	600130	Chennai Outskirts	Tamil Nadu	Very High
3	562125	Yamare	Karnataka	High
4	560064	Avalahalli	Karnataka	High (multi hotspot)
5	560049	Bommenahalli	Karnataka	High

5. Recommendations for Operations Team

01

Implement Pincode Blocking

Add the prioritised Pincodes from the final_blocked_pincodes.csv deliverable to the backend booking system. Blocking these at checkout will prevent payment capture entirely and eliminate all refund processing overhead for these locations.

02

Update Customer UI Messaging

For users attempting to order from blocked Pincodes, display a clear, friendly “Delivery currently not available in your area” message at the search or checkout stage. This sets expectations early, reduces support tickets, and maintains brand trust.

03

Review the Farmhouse / Resort Category

Consider creating a separate, scheduled delivery tier or partnership model specifically designed for farmhouses and resorts. This is a recurring source of high unmet demand that could be captured rather than blocked, if a viable delivery model is designed around it.

04

Evaluate Targeted Service Area Expansion

Clusters sitting on the immediate eastern periphery of Bengaluru and the Pimpri-Chinchwad region demonstrate high latent demand. The specific cluster coordinates can be evaluated to determine whether extending delivery radii by 3-5 kms in these pockets is financially viable thereby converting current refunds into revenue.

6. Technical Stack

The following tools and libraries were used across the end-to-end pipeline:

Category	Tool / Library
Language	Python 3.x (Jupyter Notebook)
Data Processing	pandas, numpy
Machine Learning	scikit-learn (DBSCAN clustering)
Geospatial & Mapping	folium, shapely, geopy (Nominatim API)
Visualisations	matplotlib, seaborn

This report and its deliverables are prepared as part of a data analytics competition project. All analysis was performed on real COOX booking data.